Amirpasha Hedayat

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Research Interests

My research focuses on scientific machine learning and AI for science, with an emphasis on reduced-order models, operator learning, and neural approaches for modeling complex dynamical systems. Beyond specific applications, I'm interested in how AI can serve as a general-purpose modeling tool, capable of adapting across scientific domains and levels of abstraction.

Education

PhD **University of Michigan**, Aerospace Engineering & Scientific Computing — Aug 2023 – Present
- Also affiliated with Michigan Institute for Computational Discovery and Engineering (MICDE)
- Advisor: Prof. Karthik Duraisamy

MSc **University of British Columbia**, Mechanical Engineering — Aug 2020 – Nov 2022
- GPA: 4.0/4.0 (92% local)
- Advisor: Prof. Carl Ollivier-Gooch
- Thesis: "Detecting Missing Flow Separation and Predicting Error in Drag using Supervised Machine Learning" [Link]

BS **Amirkabir University of Technology**, Aerospace Engineering — Sept 2015 – Oct 2019
- GPA: 4.0/4.0 (19.0/20.0 local)
- Advisor: Prof. Sahar Noori
- Thesis: "Estimation of Unknown Heat Flux in a 2-D Heat-Sink using Levenberg-Marquardt Inverse Algorithm"

Experience

University of Michigan, Graduate Research Assistant — Ann Arbor, MI, USA — Aug 2023 – Present
- Focus: Reduced-order Models, Scientific Machine Learning, AI for Science
- Main Tools: Python

University of British Columbia, Graduate Research Assistant — Vancouver, BC, Canada — Aug 2020 – Nov 2022
- Focus: Dimensionality Reduction, Machine learning, Error Analysis, CFD
- Main Tools: Python, C++, Matlab

Amirkabir University of Technology, Undergraduate Research Assistant — Tehran, Iran — Sep 2017 – Oct 2019
- Focus: Numerical Simulation, Inverse Problems, Heat Transfer, CFD
- Main Tools: C++, Matlab, Ansys Fluent, Pointwise

Sholeh Sanat Manufacturing and Engineering Company, Undergraduate Intern — Tehran, Iran — Jun 2018 – Aug 2018
- Focus: Design and Modification of a Heat Exchanger
- Main Tools: Autodesk Inventor, Ansys Fluent

Publications

- **Hedayat, A.**, and Ollivier-Gooch, C., "Detecting Under-Resolved Flow Physics Using Supervised Machine Learning," *AIAA Journal*, June 2023. doi: https://doi.org/10.2514/1.J062884
- **Hedayat, A.**, and Ollivier-Gooch, C., "Detecting Missing Flow Separation using Supervised Machine Learning," *AIAA SCITECH 2023 Forum*, National Harbor, MD, January 2023. doi: https://doi.org/10.2514/6.2023-1201

Talks & Posters

- “Non-intrusive Adaptive Reduced-Order Models for Predictive Modeling of Complex Systems,” *18th U.S. National Congress on Computational Mechanics (USNCCM18)*, Chicago, IL, July 2025.
- “Linear and Non-linear Reduced Dimensional Manifolds for Global Weather Predictions,” *Model Reduction and Surrogate Modeling (MORe2024)*, San Diego, CA, September 2024.
- “Detecting Missing Flow Separation using Supervised Machine Learning,” *AIAA SCITECH 2023 Forum*, National Harbor, MD, January 2023.

Skills

Programming Languages: Python, Julia, C/C++, Matlab/Octave, SQL/MySQL, HTML/CSS

AI/ML Frameworks: Pytorch, Tensorflow/Keras, Scikit-learn

Development Tools: Git/GitHub, Autotools/Make, Shell Scripting

Operating Systems: Extensive experience with MacOS, Linux (Ubuntu), and Microsoft Windows

Spoken Languages: English (proficient), Persian (native)

Awards & Honors

- Graduate Student Academic Achievement Award - UBC 2022
- Natural Sciences and Engineering Research Council of Canada (NSERC) Graduate Fellowship 2020 - 2022
- International Tuition Award - UBC 2020 - 2022
- Faculty of Applied Science Graduate Award - UBC 2020 - 2022
- The Distinguished Aerospace Engineering BSc Graduate - AUT 2019
- Graduated as the Top Student in Undergraduate Studies (out of $\sim$150) - AUT 2019
- Iran’s National Elites Foundation (INEF) Scholarship 2018 - 2019
- Department of Aerospace Engineering Academic Excellence Award - AUT 2016 - 2019